Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
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Recitation 11 Solutions

Problem 11.1

Let $X \in \{1, 2, \ldots\}$ be a random variable that represents the number of coin tosses until we observe the first heads. Suppose we repeat the experiment n times, generating values of $X_1, \ldots, X_n$ that are independent, identically distributed geometric random variables.

(a) We use the familiar parameterization of the geometric distribution

$$p_{X_i}(x; p) = (1-p)^{x-1}\, p, \qquad x = 1, 2, \ldots,$$

where $p \in (0, 1)$ is the probability of heads. Determine the maximum likelihood (ML) estimate $\hat{p}_{\text{ML}}$ of the parameter p based on the observed values $x_1^n \triangleq (x_1, \ldots, x_n)$.

Solution: Based on the distribution above, we construct

$$\begin{aligned}\ell(p; x_1^n) &= \sum_{i=1}^{n} \log p_X(x_i; p) = \sum_{i=1}^{n} (x_i - 1) \log(1-p) + n \log p \\ &= \left(\sum_{i=1}^{n} x_i - n \right) \log(1-p) + n \log p.\end{aligned}$$

We now differentiate $\ell(p; x_1^n)$ with respect to p,

$$\frac{\partial}{\partial p} \ell(p; x_1^n) = -\left(\sum_{i=1}^{n} x_i - n \right) \frac{1}{1-p} + \frac{n}{p} = \frac{n - p \sum_{i=1}^{n} x_i}{p(1-p)},$$

and setting to zero, obtain

$$\begin{aligned}\frac{\partial}{\partial p} \ell(p; x_1^n) &= 0 \\ \frac{n - p \sum_{i=1}^{n} x_i}{p(1-p)} &= 0 \\ p &= \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\frac{1}{n} \sum_{i=1}^{n} x_i}. \qquad (1)\end{aligned}$$

We verify that the first derivative of the log-likelihood function $\ell(p; x_1^n)$ changes sign from positive to negative at (1), and thus

$$\hat{p}_{\text{ML}}(x_1^n) = \frac{1}{\frac{1}{n} \sum_{i=1}^{n} x_i}.$$

(b) Now we use an alternative parameterization

$$p_{X_i}(x;\theta) = \left(1 - \frac{1}{\theta}\right)^{x-1} \frac{1}{\theta}, \qquad x = 1, 2, \ldots,$$

where $\theta \in (1, \infty)$ is the expectation of X, $\theta \triangleq \mathbb{E}[X_i] = 1/p$. Determine the ML estimate $\hat{\theta}_{\text{ML}}$ of the parameter θ based on the observed values x_1^n.

Solution: Since $\theta(p) = 1/p$ is a strictly monotonic function of p, the value of θ for which $\ell(\theta; x_1^n)$ achieves its maximum must satisfy

$$\hat{\theta}_{\text{ML}}(x_1^n) = \theta(\hat{p}_{\text{ML}}(x_1^n)) = 1/\hat{p}_{\text{ML}}(x_1^n) = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

We can verify this answer by repeating the steps of part (a) for this parameterization. Specifically,

$$\ell(\theta; x_1^n) = \sum_{i=1}^{n} \log p_X(x_i;\theta) = \left(\sum_{i=1}^{n} x_i - n\right) \log \frac{\theta - 1}{\theta} + n \log \frac{1}{\theta}$$

$$\frac{\partial}{\partial \theta}\ell(\theta; x_1^n) = \left(\sum_{i=1}^{n} x_i - n\right)\left(\frac{1}{\theta - 1} - \frac{1}{\theta}\right) - n\,\frac{1}{\theta} = \frac{\sum_{i=1}^{n} x_i - n\theta}{(\theta - 1)\theta},$$

and

$$\frac{\partial}{\partial \theta}\ell(\theta; x_1^n) = 0$$

$$\theta = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

It is straightforward to verify that the sign of the first derivative of the log-likelihood function changes from positive to negative for the value of θ above, and therefore

$$\hat{\theta}_{\text{Ml}}(x_1^n) = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

(c) Determine bias $b_{\hat{\theta}_{\text{ML}}} \triangleq \mathbb{E}\Big[\hat{\theta}_{\text{ML}}(X_1, \ldots, X_n)\Big] - \theta$ and variance $\text{var}\Big(\hat{\theta}_{\text{ML}}(X_1, \ldots, X_n)\Big)$ of the ML estimator you found in part (b).

Solution:

We proceed as follows:

$$
\begin{aligned}
b_{\hat{\theta}_{\mathrm{ML}}} &\triangleq \mathbb{E}\Big[\hat{\theta}_{\mathrm{ML}}(X_1,\ldots,X_n)\Big] - \theta \\
&= \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] - \theta \\
&= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] - \theta = 0, \qquad (2) \\
\mathrm{var}\Big(\hat{\theta}_{\mathrm{ML}}(X_1,\ldots,X_n)\Big) &= \mathrm{var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{var}(X_i) \qquad (3) \\
&= \frac{1-p}{np^2} \qquad (4) \\
&= \frac{\theta(\theta-1)}{n},
\end{aligned}
$$

where to obtain (2) we have use the linearity of expectation, to obtain (3) we have used that scaling a random variable by a constant multiplies its variance by the square of the constant and that the variance of a sum of independent random variables is equal to the sum of their variances, and to obtain (4) we have used that the variance of the geometric random variable with probability of heads p is $(1-p)/p^2 = \theta(\theta-1)$.

Note that computing expectation and variance of this estimator is relatively straightforward, while doing so for the estimator in part (a) would be quite involved. It turns out the choice of parameterization can have a significant effect on how easy or hard the subsequent computations are based on the chosen representation.

Problem 11.2

Suppose a vendor produced N widgets, numbered $1, 2, \ldots, N$. Its competitor observes $n \leq N$ widgets with numbers $X_1, X_2, \ldots, X_n$. Assume that these numbers are drawn uniformly without replacement from $\{1, 2, \cdots, N\}$.

(a) Determine the ML estimate $\hat{N}_{\mathrm{ML}}$ of N based on the observed widgets $X_1 = x_1, \ldots, X_n = x_n$.

Solution: There are $\binom{N}{n} = \frac{N!}{n!\,(N-n)!}$ possible choices of which widgets we observe.

Each of these is equally likely, so the likelihood is

$$\begin{aligned}\mathscr{L}(N; x_1^n) \triangleq p_{X_1^n}(x_1^n; N) &= \frac{1}{\binom{N}{n}} \\ &= \frac{n!\,(N-n)!}{N!} = \frac{n!}{(N-n+1)\cdot \ldots \cdot N}\end{aligned}$$

for any configuration x_1^n that represents a subset of numbers from 1 to N. It is zero for all other configurations x_1^n.

For fixed n, the likelihood function is maximized when the denominator is minimized, and therefore we want $\hat{N}_{\mathrm{ML}}$ to be as small as possible. Since x_1^n must be a subset of $\{1, 2, \cdots, N\}$, we have

$$\hat{N}_{\mathrm{ML}}(x_1^n) = \max(x_1, \ldots, x_n).$$

(b) Determine the bias of the ML estimator $b_{\hat{N}_{\mathrm{ML}}} = \mathbb{E}\Big[\hat{N}_{\mathrm{ML}}\Big] - N$. You may find this fact helpful:

$$\sum_{m=n}^{N} m\binom{m-1}{n-1} = n\binom{N+1}{n+1}.$$

Solution: We note that there are $\binom{m-1}{n-1}$ ways in which we definitely chose the widget number m and among the widgets with smaller numbers (i.e., among $m-1$ widgets) we chose $n-1$ widgets, and therefore

$$\mathbb{P}(\text{max number observed is } m) = \frac{\binom{m-1}{n-1}}{\binom{N}{n}}.$$

We now have

$$\mathbb{E}[\hat{N}_{\mathrm{ML}}(X_1^n)] = \sum_{m=n}^{N} m\,\mathbb{P}(\text{max number observed is } m) \quad = \sum_{m=n}^{N} m\frac{\binom{m-1}{n-1}}{\binom{N}{n}} = \frac{n(N+1)}{n+1},$$

where in the last step, we have used the provided hint. Thus, the bias of the ML estimator is

$$b[\hat{N}_{\mathrm{ML}}] = \mathbb{E}[\hat{N}_{\mathrm{ML}}(X_1^n)] - N = \frac{n(N+1)}{n+1} - N = -\frac{N-n}{n+1},$$

where as a reminder, note that $n \leq N$. This means that the ML estimator guesses a value of N that is lower than the true value for most configurations x_1^n. This makes sense since if we just take the maximum of the numbers on the widgets we've seen, then unless we see the widget with number N, the maximum we observe is going to be smaller than the true value of N.